The Effect of Speech Enhancement on Whisper Speech Recognition

ELEC5305 Acoustics, Speech and Signal Processing

Audio Processing Individual Project Report

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Project webpage: **https://github.com/jliu0709/elec5305-project-550284873**

Video: https://drive.google.com/file/d/14hzR9dta_tZI5xf1AQwGf2-t-4KUGKrG/view?usp=sharing

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1. Introduction

Speech enhancement is often placed before automatic speech recognition (ASR) to reduce additive noise. However, enhancement methods are usually designed for signal quality or human listening. They are not designed for the internal features of a fixed recogniser. Noise removal may help recognition, have little effect, or add new distortions. This project asks whether front-end enhancement changes the word error rate (WER) of a fixed Whisper model. It tests different noise types and input signal-to-noise ratios (SNRs).

The comparison includes noisy speech with no enhancement, power spectral subtraction, Wiener filtering, and pretrained DeepFilterNet3. Clean-speech recognition is used as a reference, and no neural model is trained. The experiment does not assume that enhancement will improve recognition, because better SNR or STOI may still be accompanied by worse WER. Compared with the earlier five-utterance experiment, this study adds STOI, Whisper log-Mel distance, corpus WER, error counts, runtime, and a larger test set.

This project uses a MATLAB-Python pipeline so that every method receives the same saved mixture. Development and test speakers are separate, parameters are chosen before the test run, and every utterance result is saved. Only the spectral-subtraction coefficient was changed to test how stronger noise removal affects residual noise, speech distortion, and WER. Some prior studies trained speech enhancement models with recognition-related objectives. This project asks whether existing enhancement methods can improve recognition when Whisper and its decoding settings are kept fixed.

2. Literature Review

Boll's spectral-subtraction method estimates a noise spectrum during non-speech and subtracts it from the noisy short-time spectrum [1]. It is computationally light but can create isolated residual components and speech distortion when the noise estimate is inaccurate. Wiener filtering instead applies a smooth frequency-dependent minimum-mean-square-error gain derived from estimated speech and noise power [2]. Both methods depend on quasi-stationary noise estimates and retain the noisy phase in the implementations used here.

Signal metrics answer different questions. Output SNR measures sample-by-sample error against an aligned clean reference. STOI estimates the intelligibility of noisy and time-frequency weighted speech. It agrees with many listening tests, but it is not designed for ASR [3]. Therefore, neither metric guarantees that Whisper will keep the speech cues it needs.

Whisper was trained on 680,000 hours of diverse weakly supervised audio and is intended to transfer to new tasks without task-specific fine-tuning [6]. This does not mean that every enhancement front end will help. DeepFilterNet uses ERB-band gains and deep filtering and is designed for real-time enhancement [7]. Previous ASR research shows that enhancement artefacts can harm a recogniser even when normal enhancement scores improve [8], [9]. For this reason, Whisper is kept fixed and WER is measured directly rather than treating cleaner audio as proof of

better recognition.

Plantinga et al. [8] trained a single-channel enhancer with a perceptual loss from a pretrained acoustic model. On Voicebank+DEMAND, their ASR-aware system reported 2.80% WER. This was more than 20% relatively lower than joint training and 14% lower than distortion-independent mask training. The result shows that enhancement can help ASR when the training objective protects recognition-relevant information.

Braun and Gamper [9] studied a real-time neural noise suppressor. They varied spectral loss terms and tested extra losses from pretrained MOS, WER, ASR and sound-event models. Their results showed a trade-off between noise reduction and speech distortion. The extra pretrained networks did not significantly outperform a strong spectral loss.

Additional work gives useful context for the methods and evaluation. Ephraim and Malah [10] used statistical short-time spectral estimation and showed the balance between noise suppression and speech-spectrum modification. Hu and Loizou [11] found that objective quality measures respond differently to enhancement distortions, which supports using several metrics rather than one score. Bagchi et al. [12] used mimic loss to make enhanced features more useful to an ASR model, while Wang et al. [13] showed that distortion-independent acoustic modelling can reduce the recognition gap caused by enhanced speech. These studies reinforce the need to measure WER directly when the recogniser is fixed.

This project uses these studies as motivation but tests a different setting. It does not train or adapt an enhancer or recogniser. Two classical methods and pretrained DeepFilterNet3 are placed before fixed Whisper base.en. The comparison covers three noise types, three SNRs and one change in spectral-subtraction strength. The experiment therefore tests front-end compatibility rather than ASR-aware enhancement training. Any agreement with [8], [9] is limited to this 50-utterance read-speech test.

LibriSpeech provides aligned 16 kHz read English speech [4], while DEMAND provides real environmental noise recordings [5]. Controlled mixtures from these datasets make it possible to compare the front ends with the same inputs. The official Whisper 80-bin log-Mel features are also compared with the matched clean reference. This distance is used only to help explain the results; a smaller value does not always mean fewer recognition errors.

These findings shaped two design choices. WER is the primary recognition metric because SNR and STOI measure different aspects of signal quality and intelligibility. All methods process the same saved noisy recordings, and Whisper settings are fixed, so changes in recognition can be compared as effects of the enhancement front end.

3. Methods

3.1 Data and experimental design

A ten-utterance Mini LibriSpeech pilot first checked the five-condition workflow at 0 dB office noise. The formal experiment used a project-specific subset of LibriSpeech dev-clean-2, not the official

LibriSpeech test set. Five development utterances from five speakers were used only for parameter selection. The independent test set contained 50 utterances from 21 speakers; no speaker appeared in both sets. Each test utterance was mixed with white, DEMAND office and DEMAND street-traffic noise at 0, 5 and 10 dB. Four primary processing conditions produced 1,800 noisy/enhanced tests, plus 50 clean recognitions. The development-selected spectral modification added 450 test outputs.

Table 1. Fixed experimental settings.

Component	Fixed setting
Audio	Mono, 16 kHz; exact length retained
Development / test	5 utterances / 50 utterances; speaker-disjoint
Noise and SNR	White, DEMAND office, DEMAND traffic; 0, 5, 10 dB
Classical STFT	512 samples; 128-sample hop; square-root periodic Hann
Noise prefix	1.0 s, excluded from recognition and all metrics
Whisper	base.en; English; temperature 0; beam size 5; FP16 off
Randomness	Saved mixture seed, channel/start and common scaling per row

3.2 Mixture construction and alignment

For clean speech s[n] and an extracted noise segment v[n], the noise gain was g = sqrt(Ps /(Pv·10^(SNR/10))). A common factor c = min(1, 0.95/max|[s; s+gv]|) was then applied to both the mixture and clean evaluation reference. This prevents clipping without changing SNR. A one-second noise-only prefix was prepended solely for noise estimation. Stored input SNR was recomputed after writing; the maximum absolute test error was 4.6×10^{-6} dB. All 2,250 test waveforms matched the reference length and had an estimated lag of zero samples.

3.3 Enhancement methods

Both classical methods used the same complex STFT X(k,m), window, hop and prefix-derived noise power estimate Pn(k). Spectral subtraction formed $\hat{P}s = \max(|X|^2 - \alpha Pn, 0.02Pn)$ and used gain $G = \min(\sqrt{\hat{P}s/|X|^2},1)$. The original method fixed α=1.0. Wiener filtering used $\hat{P}s = \max(|X|^2-Pn,0)$ and $G=\hat{P}s/(\hat{P}s+Pn)$. The noisy phase was retained and perfect-overlap synthesis divided by the accumulated squared-window weights. Relative bypass reconstruction error was below 2.2×10^{-16}.

DeepFilterNet v0.5.6 used its embedded pretrained DeepFilterNet3 model. Audio was resampled 16→48→16 kHz with polyphase filtering. The official delay-compensation option was enabled; one second of temporary tail padding was removed after inference. The final run used the official Windows x86-64 binary. No clean reference entered any enhancement method.

3.4 Recognition and metrics

Whisper base.en was fixed for every condition. Decoding used language=en, task=transcribe, temperature=0, beam size=5, condition_on_previous_text=false and the same thresholds

throughout. EnglishTextNormalizer was applied to reference and hypothesis text before jiwer alignment. Corpus WER was (S+D+I)/N using total substitutions, deletions, insertions and reference words. ΔWER is enhanced WER minus the matched noisy WER in percentage points; negative values indicate improvement. Overall uncertainty was estimated with 10,000 paired bootstrap resamples clustered by utterance. The final Python run used VS Code, Python 3.12, OpenAI Whisper 20250625, PyTorch 2.9.1 with CUDA 12.8, and an NVIDIA GeForce RTX 5070 Laptop GPU.

The additional metrics were output SNR, SNR improvement, STOI, and real-time factor (RTF). STOI was calculated with the Python pystoi implementation. The distance between official Whisper log-Mel features and the matched clean reference was also measured. Only valid speech frames were used. The log-Mel distance was used to help explain the results, not to choose a method or prove a cause.

3.5 Development-only modification

The only modification was spectral over-subtraction strength. Development candidates $\alpha \in \{0.75,1.00,1.25,1.50,2.00\}$ kept every other setting fixed. The pre-declared rule selected lowest corpus WER, then highest mean STOI, then proximity to $\alpha=1.0$. $\alpha=1.50$ achieved development WER 14.02%, compared with 14.43% for $\alpha=1.00$ and 14.54% for no enhancement. It was therefore frozen before test evaluation.

4. Results

4.1 Pilot and integrity checks

The completed ten-utterance pilot contained 258 reference words. Clean, noisy, spectral-subtraction, Wiener and DeepFilterNet3 WERs were 6.59%, 5.81%, 5.43%, 5.43% and 4.65%, respectively. Because the noisy pilot unexpectedly outperformed clean speech and the sample was small, it was used only as a pipeline check. The formal run completed all 450 mixtures, 2,300 Whisper inputs and 2,250 waveform evaluations. No missing audio, duplicate recognition key, alignment error or non-finite sample was found.

4.2 Recognition performance

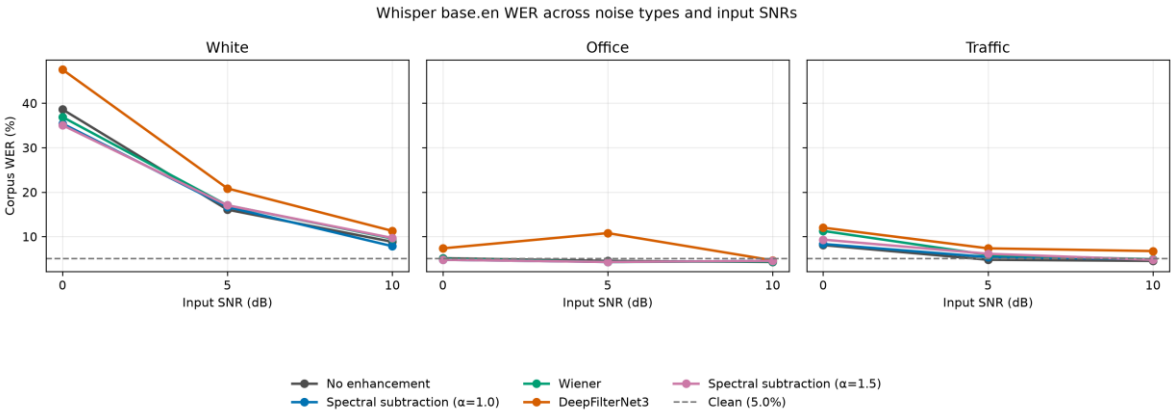


Figure 1. Corpus WER by noise type, input SNR and front-end method. The dashed line is clean-speech WER.

Table 2 summarises the overall test-set results. Clean test WER was 5.02%. Without enhancement, overall noisy WER was 10.54% (598 substitutions, 59 deletions and 117 insertions over 7,344 reference words). White noise was most difficult: unenhanced WER fell from 38.60% at 0 dB to 8.82% at 10 dB. Office and traffic conditions were much closer to the clean reference, showing that nominal SNR alone did not predict ASR difficulty.

Table 2. Overall Whisper word error rates on the test set.

Method	WER (%)	S	D	I	ΔWER (pp)	95% CI (pp)
No enhancement	10.54	598	59	117	0.00	—
Spectral subtraction α=1.0	10.24	589	55	108	−0.30	[−0.93, 0.32]
Wiener	11.03	623	57	130	+0.49	[−0.19, 1.22]
DeepFilterNet3	14.28	760	71	218	+3.74	[2.12, 5.78]
Spectral subtraction α=1.5	10.62	620	49	111	+0.08	[−0.54, 0.73]

Original spectral subtraction (α=1.0) gave the lowest overall WER, at 10.24%. However, its difference from the noisy baseline was small, and the 95% confidence interval included zero. It helped most in white noise at 0 dB (−3.19 points) and 10 dB (−0.98), while slightly worsening all traffic conditions. Wiener filtering improved white noise at 0 dB by 1.72 points but was worse overall. DeepFilterNet3 increased WER in every noise/SNR cell, with the largest change in white noise at 0 dB (+8.95 points).

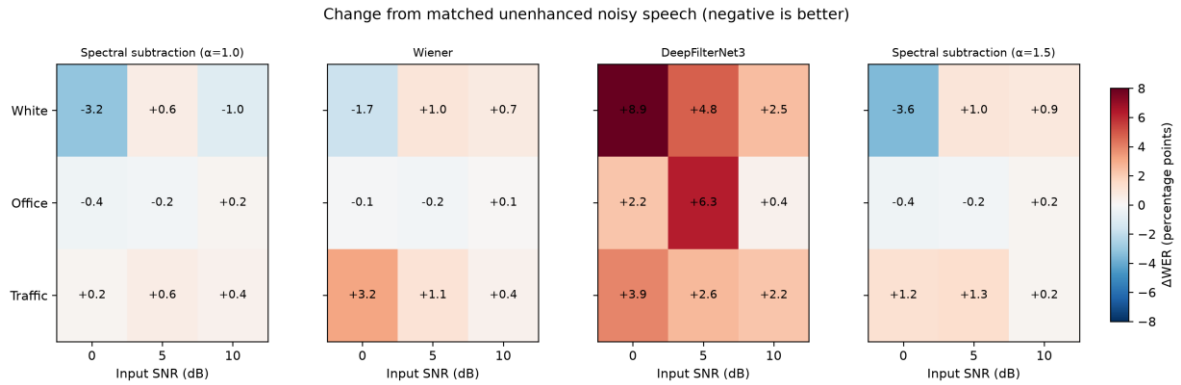


Figure 2. ΔWER relative to the matched unenhanced condition. Negative values indicate improvement.

4.3 Signal metrics versus WER

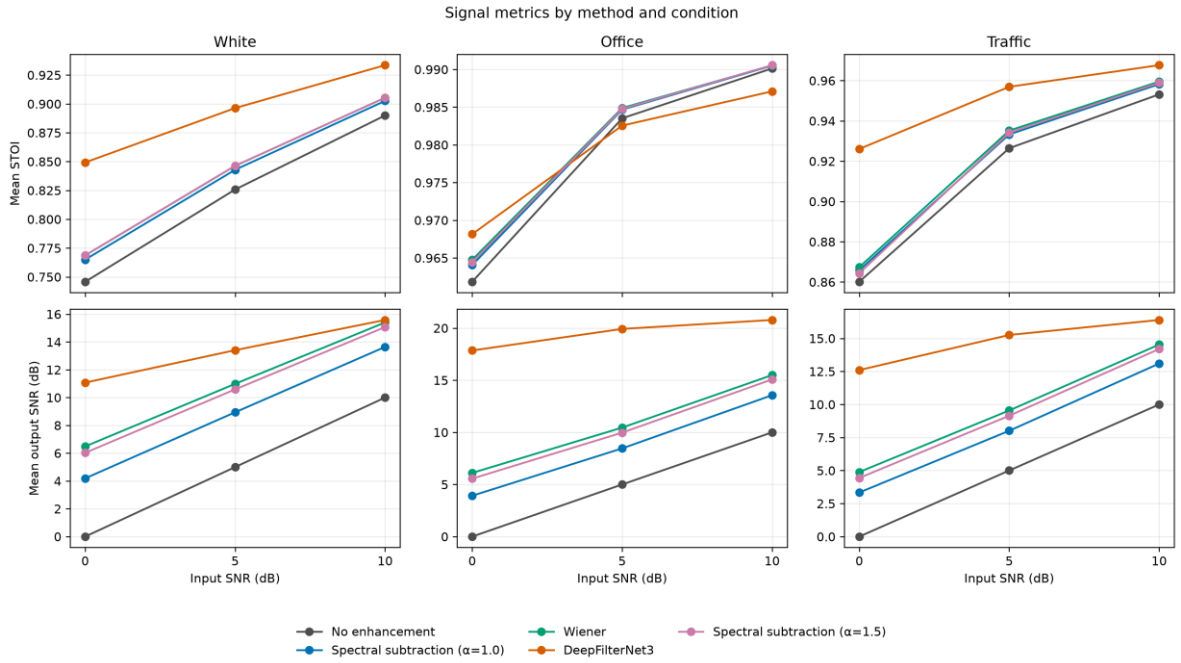


Figure 3. Mean STOI and output SNR. DeepFilterNet3 generally produced the strongest signal-level scores.

Every enhancer increased mean output SNR. DeepFilterNet3 produced about 11–21 dB output SNR. It also gave the highest STOI for white and traffic noise. However, its overall WER was 35.5% higher than the unenhanced result. For office noise at 5 and 10 dB, its STOI was 0.9826 and 0.9871. The unenhanced values were slightly higher at 0.9835 and 0.9901. These results show that better signal scores did not guarantee better Whisper results.

Across the 45 method, noise, and SNR averages, log-Mel MSE and WER had Pearson r=0.67. This shows a useful relationship, but it was not exact. DeepFilterNet3 sometimes moved the average features closer to clean while WER still increased. Whisper may therefore be sensitive to specific distortions that are hidden by an average distance.

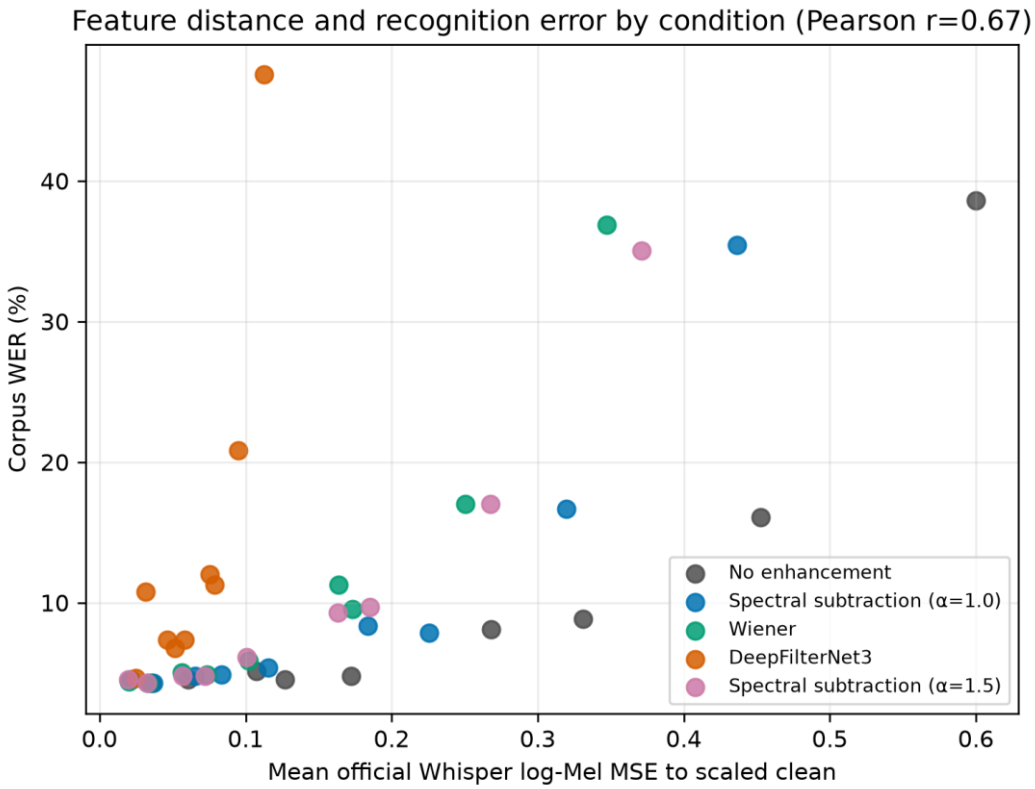


Figure 4. Mean official Whisper log-Mel distance versus corpus WER for each condition.

4.4 Modification and runtime

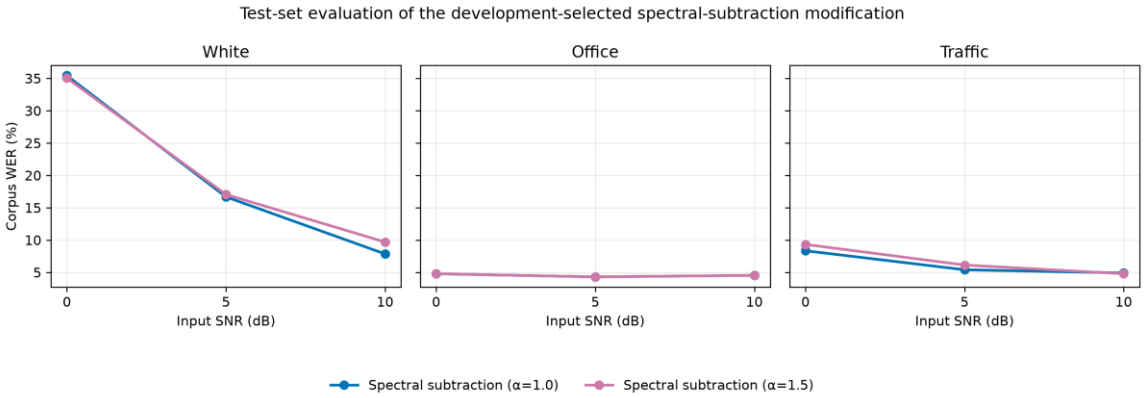


Figure 5. Independent test comparison of original α=1.0 and development-selected α=1.5 spectral subtraction.

The development set selected α=1.5 under the pre-declared rule. On the independent test set, overall WER increased from 10.24% with the original α=1.0 setting to 10.62% with α=1.5. The modified setting slightly improved white noise at 0 dB, with 35.05% WER instead of 35.42%, but it was worse at white noise at 10 dB and in all traffic conditions. Stronger subtraction improved some signal metrics, but its recognition effect changed with noise and SNR.

The median enhancement RTF was 0.0046 for spectral subtraction, 0.0044 for Wiener filtering, 0.0051 for the modified spectral subtraction, and 0.1124 for DeepFilterNet3. Whisper RTF was approximately 0.039–0.042 on the experiment computer.

4.5 Success and failure cases

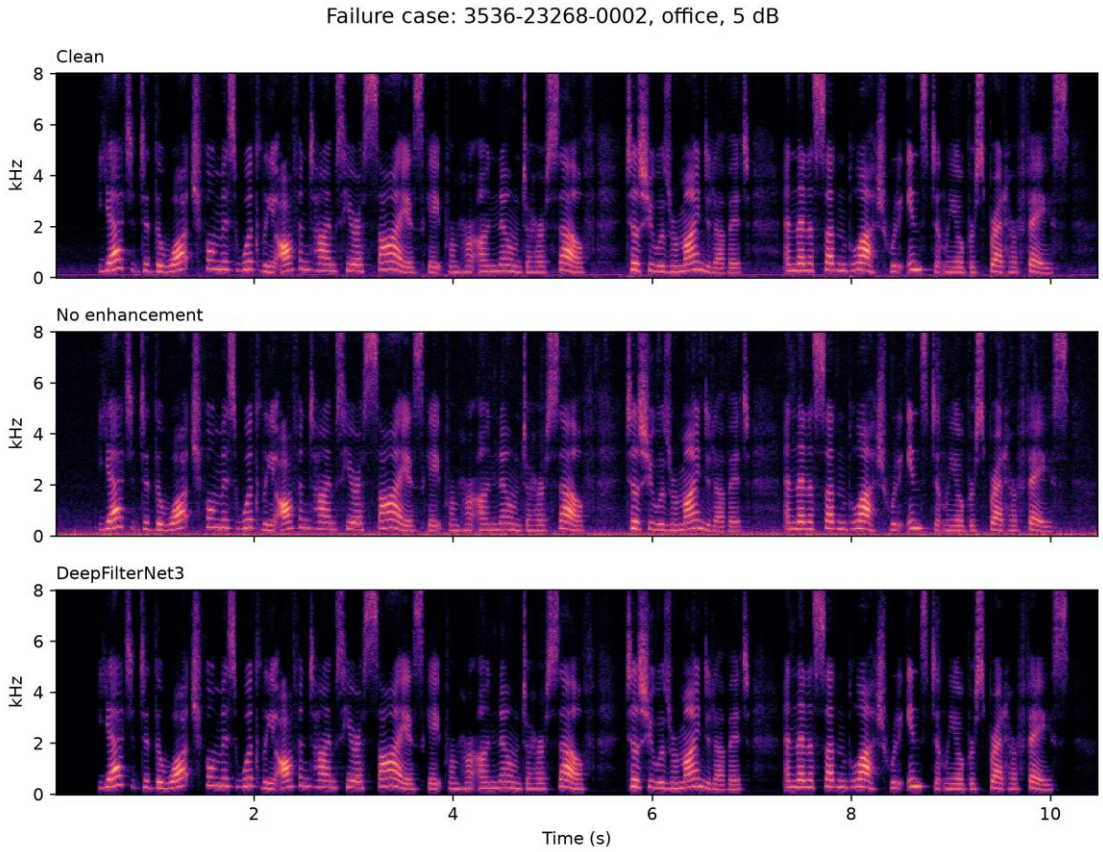


Figure 6. A failure case where DeepFilterNet3 produced a cleaner-looking spectrum but triggered repeated Whisper insertions.

For utterance 3536-23268-0002 in office noise at 5 dB, the reference ended with "then a sudden blush would instantly overspread her face." The unenhanced transcript matched the reference exactly. The DeepFilterNet3 transcript kept this passage but then added "Okay" and repeated "All right." Its 51 errors comprised 1 substitution, 0 deletions and 50 insertions. This gave 159.38% utterance WER; insertions can make WER exceed 100%. DeepFilterNet3 raised output SNR from 5.00 to 19.97 dB, while STOI fell from 0.9744 to 0.9690 and log-Mel MSE rose from 0.0209 to 0.0348. Removing this matched noisy/enhanced pair reduced the overall DeepFilterNet3 penalty from 3.74 to 3.06 percentage points. The method therefore remained worse overall.

The success case showed the opposite pattern. In white noise at 0 dB, spectral subtraction reduced the errors from nine to two. The project results include the audio and transcripts for both cases.

5. Discussion

Enhancement helped in severe white noise, where the stationary noise was close to the estimate from the one-second prefix. In office noise, spectral subtraction and Wiener filtering changed WER by less than half a percentage point at every SNR. DeepFilterNet3 behaved differently. It increased WER by 2.21 points at 0 dB and 6.25 points at 5 dB, despite improving output SNR. Its increase at

10 dB was 0.37 points.

All tested enhancers were neutral or harmful in traffic noise. For the two classical methods, the fixed initial noise estimate may not track the changing traffic noise. This explanation does not directly apply to DeepFilterNet3. Its worse WER may reflect processing distortion or a mismatch between its output and Whisper's expected input, but this experiment did not isolate the mechanism.

DeepFilterNet3 deserves particular attention because its overall WER increased from 10.54% for unenhanced noisy speech to 14.28%. In the 5 dB office-noise failure case in Figure 6, output SNR rose from 5.00 dB to 19.97 dB, but Whisper produced 50 insertion errors. Removing this matched sample reduced the overall DeepFilterNet3 penalty from 3.74 to 3.06 percentage points, so the overall degradation was not caused by this case alone. These are observed results: signal-level improvement and worse transcription occurred together. They are consistent with the general suppression-distortion concern in [8], [9], but the mechanism remains untested here. One hypothesis is that enhancement artefacts during pauses or utterance endings trigger repeated insertions. A follow-up analysis could align insertion timestamps with detected pauses and sentence endings, then compare the noisy and enhanced waveforms and Whisper log-Mel features around those locations.

The α=1.5 change did not improve the independent test result. The development set selected α=1.5, but independent-test WER increased from 10.24% for α=1.0 to 10.62%. The small development set may have contributed to an unstable parameter choice. The result also shows that stronger noise removal does not always preserve the information used by Whisper.

Several limitations restrict generalisation. The 50 utterances came from a project-specific LibriSpeech development partition and represented read English only. Only one channel per DEMAND recording and one Whisper size were tested. STOI is an intelligibility predictor rather than a listening test, and PESQ/MOS were not measured. The DeepFilterNet binary was treated as a fixed pretrained system, so architecture-specific causes of the ASR mismatch were not isolated. Finally, repeated conditions from each utterance are correlated; clustered intervals address this partly, but a larger held-out corpus would be preferable.

6. Conclusion

For the fixed Whisper base.en recogniser, enhancement helped in some conditions but not consistently. Original spectral subtraction (α=1.0) gave the lowest overall WER, 10.24% compared with 10.54% without enhancement, but the difference was small and its 95% confidence interval included zero. The development-selected α=1.5 setting increased independent-test WER from 10.24% to 10.62%, and Wiener filtering also did not improve overall WER. DeepFilterNet3 increased output SNR and improved STOI in most conditions, but it increased WER to 14.28%. Better signal scores can therefore occur together with worse recognition.

The original MATLAB experiment was extended into a complete MATLAB-Python study. The work included preparing common noisy mixtures, implementing the STFT methods, integrating

DeepFilterNet3 and Whisper, and calculating WER, STOI, SNR, log-Mel distance, and runtime. One change to spectral subtraction was also tested, and all 2,300 recognition inputs were checked. The external resources were LibriSpeech, DEMAND, OpenAI Whisper, DeepFilterNet3, jiwer, and the Python pystoi implementation. No neural model was trained. Future work could use more speakers, more Whisper sizes, more changing noise types, and a larger development set.

7. References

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